SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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Answer Script Evaluation Sheet

Assignment 1/2/3/4/5

Date:

Programme	B.E/ B. Tech	**Slot**	C
Course Code	CSA0709	**Course Name**	Computer Networks
Duration	1.30 Hrs	**Marks**	50

Q. No.	Max. Marks	Awarded Marks
Total	50	47

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ASSESSMENT TOOL 1 – COMPUTER NETWORKS

Solved Answers

TASK 1 – Size the Traffic Before Choosing Anything

1. Traffic per production line

Robot controllers: 40 × 200 bytes × 8 ÷ 0.002 s = **32 Mbps**.

Vibration sensors: 600 × 50 bytes × 8 = **0.24 Mbps**.

CCTV: 24 × 4 Mbps = **96 Mbps**.

Office users: 150 × 10% × 2 Mbps = **30 Mbps**.

Total = 32 + 0.24 + 96 + 30 = 158.24 Mbps per line.

For 8 lines: 158.24 × 8 = **1,265.92 Mbps (1.266 Gbps)**.

2. CCTV sends the most data (96 Mbps). Robot commands have the strictest delay requirement (10 ms). They are not the same.

3. One 1 Gbps uplink is sufficient for one line, but not for eight lines at peak: 1,265.92 Mbps exceeds 1 Gbps. If CCTV shares the same queue, its large packets can delay short robot commands. Use separate VLANs/queues and strict priority or time-sensitive networking for control traffic.

4. Harm from 100 ms delay (most to least): robot commands (deadline violation and unsafe control); vibration alarms (late fault detection); CCTV (live monitoring becomes delayed); office/ERP and email (inconvenience, usually tolerable).

Alternative – motion-triggered CCTV: CCTV becomes 96 × 5% = 4.8 Mbps. New line total = 32 + 0.24 + 4.8 + 30 = **67.04 Mbps**. Eight lines = **536.32 Mbps**; a 1 Gbps uplink is then sufficient, before allowing for overhead and growth.

TASK 2 – Count the Domains in the Contractor's Drawing

Original drawing: Three hubs each form one shared collision domain (3). The three hub-to-switch segments are counted as those same three domains. The switch-to-router link adds 1; router-to-second-switch adds 1; the 12 phone switch ports add 12. Total = **17 collision domains**. The router separates the two IP networks, so there are **2 broadcast domains**.

Layers: Hub – OSI Layer 1; switch – Layer 2; router – Layer 3; PC/IP phone – end devices (application through physical layers). A hub repeats bits to every port and does not examine MAC addresses, so devices share bandwidth and collisions.

Connections: Hub ports are multipoint/shared; switch-to-router and individual switch-to-device links are point-to-point.

Standards:

Technology	Standard	Defining body
Ethernet	IEEE 802.3	IEEE
Wi-Fi	IEEE 802.11	IEEE
IPv4	RFC 791	IETF
OSI model	ISO 7498	ISO
5G	5G specifications / IMT-2020	3GPP / ITU

Standard-compliant equipment improves interoperability, safety, predictable performance, support and future replacement.

Alternative – replace hubs with three switches: each PC port becomes its own collision domain. With 24 PCs, three switch uplinks, the router links and 12 phone ports, there are approximately **41 collision domains**. Broadcast domains remain **2** because ordinary switches forward broadcasts; VLANs or a router are needed to divide them. Switching removes shared-hub collisions but does not stop broadcasts.

TASK 3 – Read a Captured Frame Layer by Layer

Ethernet II: Layer 2, frame. IPv4: Layer 3, packet. TCP: Layer 4, segment. Modbus command: application data (Layer 7).

TTL changes from 64 to 61, so the packet crossed **3 routers**.

For the 12-byte command: Ethernet header 14 + IPv4 header 20 + TCP header 20 + data 12 + FCS 4 = **70 bytes**. Overhead = 58 bytes; 58/70 × 100 = **82.86%**.

For a 1,400-byte camera payload: total = 14 + 20 + 20 + 1,400 + 4 = **1,458 bytes**. Overhead = 58/1,458 × 100 = **3.98%**. Small commands have a much higher overhead percentage.

Fault mapping: wrong MAC address – Data Link layer; TTL reaches zero – Network layer; closed destination port – Transport layer.

TCP/IP mapping: Application = OSI Layers 5–7; Transport = Layer 4; Internet = Layer 3; Network Access/Link = Layers 1–2.

Code (send function):

```
def send(message):
    pdu = message
    for layer, tag in LAYERS:
        if tag == "ETH":
            pdu = f"[{tag}|{pdu}|FCS]"
        else:
            pdu = f"[{tag}|{pdu}]"
        print(f"{layer} adds {tag} -> {len(pdu)} bytes")
    return pdu
```

The supplied sample's displayed lengths are based on its string-format headers. For the industrial Ethernet alternative, removing IPv4 (20 bytes) and TCP (20 bytes) saves **40 bytes per command**. At 100 Mbps, 40 × 8 = 320 bits saved per command; 100,000,000/320 = **312,500 commands/s** of equivalent link capacity. The plant loses IP routing, TCP reliability/ordering and compatibility with ordinary IP networks.

TASK 4 – Choose a Medium for Each Run

(a) Robot cell, 85 m beside welders: **Cat6A S/FTP** shielded copper, correctly grounded and installed, to resist EMI; fibre is even more immune if isolation is required. (b) Office to shop-floor switch, 140 m: **multimode fibre**, because copper Ethernet channel limits are normally 100 m. (c) Shop floor to data centre, 550 m: **multimode fibre** for this campus distance. (d) Plant to head office, 400 km: **single-mode fibre/leased line** through a carrier. Copper Ethernet cannot span 400 km.

Shannon: C = B logI(1 + SNR). At 40 dB, SNR = 10,000; C = 250 MHz × logI(10,001) = **3,321.93 Mbps**. At 15 dB, SNR = 31.62; C = 250 MHz × logI(32.62) = **1,257.5 Mbps**. The noisy weld area has lower theoretical capacity and less margin; shielded cable/fibre and robust industrial equipment are preferred.

Actual speed is lower than Shannon capacity because of coding gaps, interference, noise, attenuation, protocol overhead, hardware limits and implementation constraints.

Code:

```
def nyquist(bandwidth_hz, levels):
    return 2 * bandwidth_hz * math.log2(levels)

def shannon(bandwidth_hz, snr_db):
    snr = 10 ** (snr_db / 10)
    return bandwidth_hz * math.log2(1 + snr)
```

Telephone line at 56 kbps: required SNR = 2^(56,000/3,100) − 1 ≈ 2,577, or 34.1 dB. This is demanding for a real telephone channel; practical modem performance is constrained by channel quality and network conversion.

Wi-Fi 6 at 160 MHz and 15 dB: C = 160 MHz × logI(1 + 31.62) **≈ 804.8 Mbps**. This theoretical rate does not guarantee a 10 ms deadline; contention, interference and roaming make Wi-Fi unsuitable as the sole safety-critical robot-control link.

TASK 5 – Wi-Fi 6 or 5G for Moving Equipment?

Trip time = 1,200 m / 2 m/s = **600 s**. Wi-Fi AP spacing 30 m gives about 1,200/30 = 40 cells and approximately **39 handovers** between cells; interruption = 39 × 150 ms = **5.85 s**. A 300 m 5G cell gives 4 cells and about **3 handovers**; interruption = 3 × 20 ms = **60 ms**. (Boundary conventions can change the count by one.)

5G service mapping: AGVs and robot control – URLLC; CCTV – eMBB; large numbers of low-rate sensors – mMTC.

Bluetooth suits short-range peripherals; Zigbee suits low-power mesh sensors; LoRaWAN suits sparse, small, long-range telemetry. Wi-Fi/5G are better for high-rate data.

802.11r fast roaming can reduce Wi-Fi handover time, but cannot guarantee a 10 ms end-to-end deadline under interference, congestion and variable backhaul delay.

LoRaWAN alternative: 600 × 50 bytes/s × 8 = **240,000 bps = 240 kbps**, far above a 5 kbps gateway capacity. Only a small subset of low-frequency sensors can use one such gateway; high-rate or frequent sensors must remain on a higher-capacity network.

TASK 6 – Interpret an Ethernet Port Report

A late collision occurs after the normal collision window and usually indicates a duplex mismatch or cabling/segment problem. CRC errors mean received frames fail the frame-check sequence. Together with 100 Mbps half-duplex and a gigabit switch set to auto, a **duplex mismatch** is likely.

Correct both ends to auto-negotiation, or explicitly set both ends to the same speed and full duplex. Linux example: sudo ethtool -s eth0 speed 1000 duplex full autoneg on (use a supported setting; preferably enable auto on both ends).

01:00:5e:0a:01:02 is an IPv4 multicast Ethernet MAC address (group, not one device). The first three bytes 3C:52:82 are the OUI, identifying the registered manufacturer of the interface.

Jumbo frames can improve large CCTV-storage transfers by reducing per-packet overhead, but only if every device and link in the path (NICs, switches, storage server and configuration) supports the same MTU. They do not fix a slow or faulty link.

A fast-recovery industrial ring can reroute after one cable cut; recovery is typically tens of milliseconds depending on protocol/configuration. A star has no alternate path: devices on the failed branch lose connection until repair. The ring maintains service for devices reachable through the alternate direction.

TASK 7 – Will the Receiver Notice the Error?

Generator 1011 has degree 3, so append 000 to the data 1101011011 and divide by 1011 using XOR. The remainder is **100**; transmitted codeword = **1101011011100**. A received frame with non-zero remainder is rejected; 1101011011101 has remainder 001 and is rejected.

A degree-3 CRC detects every burst error of length **≤** 3 bits. It can miss an error pattern that is a multiple of the generator, for example the generator pattern itself **1011** (aligned within a longer frame), because its remainder is zero.

Two-dimensional parity example (even parity): rows 1010, 0110, 1100, 0011. Add one parity bit to each row and one parity row to make every row and column even. It detects any single-bit error and many multiple-bit errors, but a four-corner rectangle (errors at the four corners of two selected rows and two selected columns) leaves all row and column parities unchanged.

Internet checksum: words 4500 + 0073 + 0000 + 4000 = 8573 (hex). One’s complement = **7A8C**.

Modulo-2 division code:

```
def mod2_div(dividend, divisor):
    rem = list(dividend)
    for i in range(len(dividend) - len(divisor) + 1):
        if rem[i] == '1':
            for j in range(len(divisor)):
                rem[i+j] = str(int(rem[i+j]) ^ int(divisor[j]))
    return ''.join(rem[-(len(divisor)-1):])
```

Even parity catches every odd number of flipped bits, but misses even-numbered errors (e.g. 2-bit and 4-bit errors). CRC-3 detects all single-bit errors and all bursts up to 3 bits, but can miss some longer patterns that are multiples of 1011.

TASK 8 – Correct the Error or Ask Again?

For m = 16, choose r such that 2I **≥** m + r + 1. r = 5 because 32 **≥** 22. Total code length = 21 bits; overhead = 5/16 × 100 = **31.25%** relative to data.

For codewords {00000, 01011, 10101, 11110}, pairwise minimum Hamming distance is **3**. It detects up to 2 errors and corrects up to 1 error.

Received 0110111 differs from valid 0110011 at bit 5. Corrected codeword = **0110011**; data bits at positions 3,5,6,7 = **1011**.

For a noisy radio link with 40 ms retransmission delay, FEC is preferable for time-critical readings because it corrects errors without waiting. ARQ is useful when delay is acceptable and retransmission is reliable. Video and 5G use FEC to reduce delay and maintain continuous delivery.

With two flipped bits, a single-error-correcting Hamming decoder may misinterpret the syndrome as a single-bit error and flip a third bit, producing an incorrect result; it cannot reliably correct double-bit errors.

TASK 9 – How Big Must the Window Be?

Frame size = 1,500 × 8 = 12,000 bits. Transmission time = 12,000/10I = **12 µs**. One-way propagation = 400,000/(2×10I) = **2 ms**. a = 2 ms/12 µs = **166.67**.

Stop-and-wait utilisation = 1/(1+2a) = 1/334.33 = **0.299%**, giving about **2.99 Mbps** on a 1 Gbps link.

Minimum window for full utilisation = ceiling(1+2a) = **335 frames**. Go-Back-N needs k bits where 2I−1 ≥ 335, so **9 bits**. Selective Repeat needs 2I ≥ 2×335, so **10 bits**.

Selective Repeat resends only lost frames; Go-Back-N resends the lost frame and subsequent unacknowledged frames. Selective Repeat needs more receiver buffering and per-frame tracking.

HDLC and PPP use a flag byte 01111110 (0x7E) to mark frame boundaries; bit/byte-stuffing prevents the flag pattern appearing as data.

Selective Repeat window 64: U = 64/(1+2×166.67) = **19.10%**. Throughput = 0.1910 × 1 Gbps = **191 Mbps**. Receiver buffer = 64 × 1,500 = **96,000 bytes** (about 96 kB).

TASK 10 – Find Where the Delay Comes From

Expected one-way propagation for 400 km fibre = 400,000/(2×10I) = **2 ms**. This is the minimum physical transit component; the observed destination RTT also includes return propagation, routing, queuing and processing. A 4 ms round-trip propagation baseline is expected.

A single 38.2 ms traceroute response at hop 3 can be ICMP rate-limiting or temporary queuing/CPU delay at that router; it does not prove forwarding traffic is delayed by 38.2 ms. Stars at hop 5 mean no response to the probe, often due to filtering or rate-limiting, not necessarily a fault.

From the displayed results, hop 3 has the largest isolated spike. The sustained end-to-end increase should be judged from later hops and repeated measurements, not one probe.

Useful commands: traceroute server (Linux/macOS), tracert server (Windows), ping server, and Wireshark capture with Ethernet, IP, TCP/UDP and application headers inspected.

SNMP counters and NetFlow provide continuous interface/error/traffic trends and can reveal congestion or loss between tests. Occasional ping/traceroute is simple but only samples conditions when run.

TASK 11 – Defend the Network Design

Recommended design: Cat6A S/FTP for short, electrically noisy robot-cell connections; fibre for runs over 100 m and the plant backbone; single-mode fibre/leased carrier service for the 400 km head-office link. Use managed industrial switches, VLAN separation, QoS and redundant paths.

Use wired Ethernet/industrial Ethernet with priority scheduling for fixed robot controllers; private 5G URLLC for mobile AGVs; Wi-Fi 6 for office devices and non-critical mobility; eMBB for cameras; mMTC/LoRaWAN for low-rate sensors where the capacity budget permits.

Use CRC/FCS on Ethernet, FEC on time-critical wireless links, and ARQ/TCP where retransmission delay is acceptable. Use sliding-window transport on long-distance links; at 400 km, stop-and-wait gives only about 0.3% utilisation, while a 335-frame window can fill the 1 Gbps link.

Two major remaining risks are radio interference/coverage or handover failures, and backbone/provider outages or congestion. Five plants require an appropriately sized WAN: at the original traffic rate, five plants × 8 lines × 158.24 Mbps = **6.3296 Gbps** before overhead. Use multiple 10 Gbps or higher links, traffic engineering and local processing.

Private 5G instead of all wired: use 5G for AGVs and selected mobile devices; keep fixed robots, high-volume CCTV aggregation, switches and servers on wired Ethernet/fibre. This avoids putting critical traffic and 1.266 Gbps per-plant peak traffic entirely on a shared radio resource.

TASK 12 – Re-evaluate and Adapt the Design

A. Impact and traffic: The 50 AGVs add traffic not quantified in the question, so an exact new Mbps total cannot be calculated without each AGV's command size and reporting frequency. If each AGV uses the same 200-byte command every 2 ms as a robot controller, added traffic = 50 × 200 × 8/0.002 = **40 Mbps**. New line-equivalent plant traffic = 1,265.92 + 40 = **1,305.92 Mbps** (assuming the 50 AGVs are additional to the eight lines).

B. Original design test: Wi-Fi with 39 handovers at 150 ms produces about 5.85 s total interruption per 600 s trip. 5G with about 3 handovers at 20 ms produces about 60 ms total interruption. Both totals exceed a 10 ms command deadline if

interruption coincides with a command; therefore the original ordinary-Wi-Fi design is not acceptable for guaranteed AGV control.

C. Adaptation: MODIFY the design. Deploy private 5G with URLLC and overlapping cells for AGVs; keep fixed robot control on wired industrial Ethernet; retain Wi-Fi 6 for office traffic; use fibre for the backbone and CCTV aggregation. Add redundant coverage, QoS, local edge control and tested failover.

D. Defence: The 5G handover estimate (20 ms) is much shorter than Wi-Fi's 150 ms, but still exceeds the 10 ms command deadline. Therefore use local autonomous safety logic and seamless redundant coverage so a handover does not interrupt essential control. Do not claim a radio handover alone guarantees zero missed commands.

ADDITIONAL QUESTIONS VISIBLE ON PAGES 12–13

The scan includes a partial preceding question whose heading and numerical data are missing. The visible final instruction asks to plot Reno and BBR traces and count extra Reno segments; that part cannot be solved accurately without the missing setup and trace data.

Q3 – Why Does the Home Page Take 8 Seconds?

There are 81 objects total (one HTML file + 80 objects), with RTT = 50 ms and negligible transmission time.

Non-persistent HTTP, one connection at a time: approximately 2 RTT per object (one to establish TCP and one request/response) = 81 × 100 ms = **8.1 s**.

Persistent HTTP/1.1 without pipelining: approximately 1 RTT for TCP setup + 81 RTT for sequential requests = **4.1 s**. With pipelining: 1 RTT setup + 1 RTT for the HTML request + 1 RTT for the pipelined objects **≈ 0.15 s** (simplified model).

Six parallel non-persistent connections: requests occur in about ceil(81/6)=14 batches; approximate time = 14 × 2 RTT = **1.4 s** (ignoring browser scheduling and connection reuse).

HTTP/2 multiplexes streams over one TCP connection, but TCP loss blocks delivery of later bytes until the missing segment is recovered (TCP head-of-line blocking). HTTP/3 uses QUIC over UDP, so loss in one stream does not block unrelated streams.

Cache-Control can allow fresh objects to be reused without a network request. ETag supports conditional requests; unchanged objects return 304 Not Modified, avoiding full object transfer. If all 80 objects are fresh in cache, the second visit may need only the main HTML or can be almost entirely local.

Ranked improvements: (1) HTTP/2 or HTTP/3 multiplexing and fewer round trips; (2) caching with Cache-Control/ETag; (3) bundle many tiny objects or reduce object count.

Experimental investigation: Actual connection counts, h1/h2/h3 protocol, load times at 0/50/200 ms added delay and cache results require a real browser/network test; they cannot be inferred from the question.

Alternative – bundle 80 objects into 8: total objects = 9 including HTML. Non-persistent sequential **≈** 9 × 100 ms = **0.9 s**. Persistent without pipelining **≈** 1 RTT setup + 9 RTT = **0.5 s**. Persistent with pipelining **≈** 0.15 s. Bundling greatly improves non-persistent HTTP by reducing request count; HTTP/2 also reduces request/connection overhead.

Q4 – How Many Queries Reach the University's DNS Server?

For one resolver, query rate **≈** 3,600/TTL. At TTL 300 s: **12 queries/hour**. At TTL 30 s: **120 queries/hour**. Across R resolvers, multiply by R (assuming each resolver has steady demand and cache expiry).

The 150,000 students are hidden behind ISP recursive resolvers. A resolver can answer many students from its cache, so the authoritative server mainly sees cache misses from resolvers. More resolvers and shorter TTLs increase authoritative queries.

A phone normally sends a recursive query to its ISP resolver. The resolver performs iterative queries: root server → .in TLD server → ac.in delegation → simats.ac.in authoritative server, then returns and caches the answer.

A: IPv4 address; AAAA: IPv6 address; CNAME: alias; MX: mail exchangers; NS: authoritative name servers; TXT: verification and policy text (such as SPF). DNS usually uses UDP for low overhead and speed; TCP is used for large/truncated responses and zone transfers.

A very long TTL can leave clients using an old address after a server move. Lower the TTL in advance, wait for the previous TTL to expire, move the service, verify it, then increase the TTL.

DNS code completion:

```
def resolve(name):
    if name in cache:
        return cache[name], 0
    server, queries = "root", 0
    while True:
        queries += 1
        matches = [(suffix, value) for suffix, value in SERVERS[server].items()
                   if name.endswith(suffix)]
        if not matches:
            raise ValueError("No delegation for " + name)
        suffix, value = max(matches, key=lambda item: len(item[0]))
        if any(ch.isdigit() for ch in value):
            cache[name] = value
            return value, queries
        server = value
```

Recommendation: Use a moderate TTL (for example, 300 seconds during normal operation, reduced before a planned migration), multiple authoritative DNS servers, monitoring and tested resolver capacity. At TTL 300 s each resolver contributes about 12 lookups/hour; at TTL 30 s it contributes 120/hour, a tenfold increase.